

Telecom Churn Dataset: Data Dictionary & Predictive Analysis

Comprehensive Column-by-Column Architectural & Machine Learning Guide

Dataset Domain: Telecom & Subscription Analytics
Total Features: 21 Columns

Target Variable: Churn (Binary)

DEMOGRAPHICS

4 Features

Gender, Senior, Partner, Dependents

ACCOUNT & TENURE

5 Features

Tenure, Contract, Payment, Billing

SERVICES ENROLLED

9 Features

Phone, Internet, Security, Streaming

FINANCIAL METRICS

2 Features

Monthly & Total Charges

1. EXECUTIVE SCHEMA OVERVIEW

Quick-reference mapping of all 21 dataset features, categorizing data types, business criticality, and predictive power for customer churn modeling.

COLUMN NAME	DATA TYPE	CATEGORY	IMPORTANCE	PREDICTS CHURN?
customerID	String / Identifier	Identity	Low	No (ID Only)
gender	Categorical (Binary)	Demographic	Low	No (Negligible)
SeniorCitizen	Binary Integer (0/1)	Demographic	Medium	Yes (Moderate)
Partner	Categorical (Binary)	Demographic	Medium	Yes (Moderate)
Dependents	Categorical (Binary)	Demographic	Medium	Yes (Moderate)
tenure	Numeric Integer	Account	Critical	Yes (Strong Inverse)
PhoneService	Categorical (Binary)	Services	Low	No (Weak)
MultipleLines	Categorical (3-Level)	Services	Low	Slightly
InternetService	Categorical (3-Level)	Services	High	Yes (Strong)
OnlineSecurity	Categorical (3-Level)	Add-on Services	High	Yes (Strong Inverse)
OnlineBackup	Categorical (3-Level)	Add-on Services	Medium	Yes (Moderate)
DeviceProtection	Categorical (3-Level)	Add-on Services	Medium	Yes (Moderate)
TechSupport	Categorical (3-Level)	Add-on Services	High	Yes (Strong Inverse)
StreamingTV	Categorical (3-Level)	Add-on Services	Medium	Yes (Moderate)
StreamingMovies	Categorical (3-Level)	Add-on Services	Medium	Yes (Moderate)
Contract	Categorical (3-Level)	Account / Terms	Critical	Yes (Primary Predictor)

COLUMN NAME	DATA TYPE	CATEGORY	IMPORTANCE	PREDICTS CHURN?
PaperlessBilling	Categorical (Binary)	Billing	Medium	Yes (Moderate)
PaymentMethod	Categorical (4-Level)	Billing	High	Yes (Strong)
MonthlyCharges	Numeric Float	Financial	High	Yes (Strong Positive)
TotalCharges	Numeric Float	Financial	High	Yes (Strong)
Churn	Categorical / Binary	Target Variable	Critical	Target Label

2. DETAILED COLUMN ANALYSIS

Comprehensive architectural analysis of each feature in the telecom churn dataset, covering business meaning, purpose, machine learning importance, and churn predictive dynamics.

1. customerID

STRING / UNIQUE KEY

LOW IMPORTANCE

PREDICTS: NO

Business Meaning:

Unique alphanumeric string identifying an individual telecom subscriber (e.g., "7590-VHVEG").

Business Purpose:

Acts as the primary key to merge customer data across billing, support, CRM, and usage databases without duplicate entries.

Can it predict churn? No.

Unique primary keys carry high cardinality and zero generalizable statistical signal. Including it in predictive models will cause severe overfitting. It must be dropped or set as an index prior to model training.

2. gender

CATEGORICAL (BINARY)

LOW IMPORTANCE

PREDICTS: NO

Business Meaning:

Biological sex or legal gender identity of the account owner ("Male", "Female").

Business Purpose:

Used for demographic segmentation, customer persona profiling, and targeted marketing campaigns.

Can it predict churn? No (Negligible).

Churn distribution between male and female subscribers in standard telecom benchmarks shows near-identical churn rates (~26% for both). It provides minimal split-criterion value in tree-based algorithms.

3. SeniorCitizen

BINARY INTEGER (0 / 1)

MEDIUM IMPORTANCE

PREDICTS: YES

Business Meaning:

Flag indicating whether the subscriber is aged 65 or older (1 = Senior Citizen, 0 = Non-Senior).

Business Purpose:

Helps tailor customer care, accessibility offerings, product packaging, and fixed-income pricing strategies.

Can it predict churn? Yes (Moderate).

Senior citizens exhibit significantly higher churn rates (~41.6%) compared to non-seniors (~23.6%). This elevated churn is often linked to fixed budgets, tech support usability issues, and digital onboarding friction.

4. Partner

CATEGORICAL (BINARY)

MEDIUM IMPORTANCE

PREDICTS: YES

Business Meaning: Indicates whether the customer lives with a spouse or partner ("Yes", "No").

Business Purpose: Measures household stability and multi-user potential, enabling multi-line and family plan upsell opportunities.

Can it predict churn? Yes (Moderate Inverse). Customers with a partner have lower churn (~20%) than single subscribers (~33%). Joint account decisions and household service integration create higher switching barriers.

5. Dependents

CATEGORICAL (BINARY)

MEDIUM IMPORTANCE

PREDICTS: YES

Business Meaning: Indicates whether the subscriber has financial dependents like children or elderly parents ("Yes", "No").

Business Purpose: Identifies family households for bundle offerings, parental control services, and long-term retention contracts.

Can it predict churn? Yes (Moderate Inverse). Customers with dependents are substantially less likely to churn (~15.5% churn vs. ~31.3% for those without). Family disruption costs make switching providers undesirable.

6. tenure

NUMERIC INTEGER (MONTHS)

CRITICAL IMPORTANCE

PREDICTS: YES (STRONG)

Business Meaning: The total number of months the customer has stayed with the telecom company.

Business Purpose: Measures customer lifetime duration, brand loyalty, cohort survival, and historical account maturity.

Can it predict churn? Yes (Strong Inverse Relationship). Tenure is one of the top 3 predictors in telecom churn models. Churn risk is highest in months 1–12 (new customer attrition) and drops exponentially as tenure reaches 24+ months.

7. PhoneService

CATEGORICAL (BINARY)

LOW IMPORTANCE

PREDICTS: NO (WEAK)

Business Meaning: Indicates if the customer subscribes to landline or voice telephone services ("Yes", "No").

Business Purpose: Identifies core Voice service adoption and helps separate internet-only customers from traditional telephony users.

Can it predict churn? No / Weak. Over 90% of telecom customers subscribe to phone service, making its univariate predictive power minimal. However, it is necessary to contextualize `MultipleLines`.

8. MultipleLines

CATEGORICAL (3-LEVEL)

LOW IMPORTANCE

PREDICTS: SLIGHTLY

Business Meaning: Indicates whether the customer has multiple telephone lines ("Yes", "No", "No phone service").

Business Purpose: Evaluates voice service depth, revenue per account (ARPU), and multi-device household utilization.

Can it predict churn? Slightly. Customers with multiple lines have a slightly higher churn rate than single-line users, primarily due to higher monthly bills, but its isolated predictive weight is low compared to contract type.

9. InternetService

CATEGORICAL (3-LEVEL)

HIGH IMPORTANCE

PREDICTS: YES (STRONG)

Business Meaning: Type of internet connectivity subscribed ("DSL", "Fiber optic", "No").

Business Purpose: Core revenue generator and technology classifier; determines network bandwidth and service delivery channel.

Can it predict churn? Yes (Strong Predictor). Fiber optic subscribers exhibit significantly higher churn rates (~42%) than DSL users (~19%) or non-internet users (~7%). Fiber services carry higher prices and competitive market churn options.

10. OnlineSecurity

CATEGORICAL (3-LEVEL)

HIGH IMPORTANCE

PREDICTS: YES (STRONG)

Business Meaning: Indicates subscription to cybersecurity add-ons like antivirus and phishing protection ("Yes", "No", "No internet service").

Business Purpose: High-margin value-added service (VAS) designed to enhance network trust and reduce security-related complaints.

Can it predict churn? Yes (Strong Inverse). Customers WITHOUT online security churn at ~41%, whereas those WITH security add-ons churn at only ~14%. Security services serve as a key retention anchor.

11. OnlineBackup

CATEGORICAL (3-LEVEL)

MEDIUM IMPORTANCE

PREDICTS: YES

Business Meaning: Subscription to cloud storage backup services for data preservation ("Yes", "No", "No internet service").

Business Purpose: Increases customer sticky factors by storing personal files and documents on carrier infrastructure.

Can it predict churn? Yes (Moderate Inverse). Enrolled customers churn at ~21.5% versus ~39.9% for non-enrolled users. Cloud data storage increases user switching costs due to migration friction.

12. DeviceProtection

CATEGORICAL (3-LEVEL)

MEDIUM IMPORTANCE

PREDICTS: YES

Business Meaning: Hardware warranty/insurance plan for routers, modems, and mobile devices ("Yes", "No", "No internet service").

Business Purpose: Mitigates customer dissatisfaction arising from equipment failure or accidental device damage.

Can it predict churn? Yes (Moderate Inverse). Subscribers without hardware protection churn at ~39%, whereas protected subscribers churn at ~22.5%. Hardware replacement issues are a primary trigger for operational churn.

13. TechSupport

CATEGORICAL (3-LEVEL)

HIGH IMPORTANCE

PREDICTS: YES (STRONG)

Business Meaning: Premium priority technical support helpline access ("Yes", "No", "No internet service").

Business Purpose: Reduces technical frustration, ticket resolution times, and call center escalation friction.

Can it predict churn? Yes (Strong Inverse). Lack of technical support yields a high churn rate (~41.6%), whereas subscribers with dedicated tech support churn at only ~15.2%. Technical resolution is a major loyalty driver.

14. StreamingTV

CATEGORICAL (3-LEVEL)

MEDIUM IMPORTANCE

PREDICTS: YES

Business Meaning: Subscription to provider-partnered television streaming channels ("Yes", "No", "No internet service").

Business Purpose: Drives content ecosystem engagement and increases overall monthly recurring revenue (ARPU).

Can it predict churn? Yes (Moderate). While streaming services increase overall bill size (which correlates with higher churn risk), when combined with long tenure, streaming acts as an engagement retention vehicle.

15. StreamingMovies

CATEGORICAL (3-LEVEL)

MEDIUM IMPORTANCE

PREDICTS: YES

Business Meaning: Subscription to premium movie streaming services ("Yes", "No", "No internet service").

Business Purpose: Entertainment add-on for digital consumption households, increasing bandwidth usage and ARPU.

Can it predict churn? Yes (Moderate). Behaves similarly to `StreamingTV`. Customers with active entertainment packages churn at ~30% when unbundled, but churn drops when integrated into multi-service contracts.

16. Contract

CATEGORICAL (3-LEVEL)

CRITICAL IMPORTANCE

PREDICTS: YES (PRIMARY)

Business Meaning: Contractual commitment term length ("Month-to-month", "One year", "Two year").

Business Purpose: Establishes revenue predictability, customer lock-in duration, and legal termination penalty terms.

Can it predict churn? Yes (Single Strongest Feature). Month-to-month contract holders have a massive ~42.7% churn rate, compared to 11.3% for One-year and only ~2.8% for Two-year contracts. It is the dominant feature in feature importance evaluations.

17. PaperlessBilling

CATEGORICAL (BINARY)

MEDIUM IMPORTANCE

PREDICTS: YES

Business Meaning: Indicates whether statements are sent via electronic billing rather than physical mail ("Yes", "No").

Business Purpose: Reduces postal billing operational costs and aligns with digital customer touchpoint strategies.

Can it predict churn? Yes (Moderate Positive). Paperless billing subscribers churn at ~33.6%, whereas non-paperless users churn at ~16.3%. Digital billing users are typically more tech-savvy and price-sensitive, facilitating easier switching to competitors.

18. PaymentMethod

CATEGORICAL (4-LEVEL)

HIGH IMPORTANCE

PREDICTS: YES (STRONG)

Business Meaning: Billing transaction mechanism ("Electronic check", "Mailed check", "Bank transfer (automatic)", "Credit card (automatic)").

Business Purpose: Manages payment processing fees, revenue collection efficiency, and payment failure risks.

Can it predict churn? Yes (Strong Predictor). Electronic check users have an extraordinarily high churn rate (~45.3%), whereas automatic payment methods (Credit Card / Bank Transfer) average ~16% churn. Electronic check users experience active monthly friction.

19. MonthlyCharges

NUMERIC FLOAT (CURRENCY)

HIGH IMPORTANCE

PREDICTS: YES (STRONG)

Business Meaning: The current monthly fee billed to the customer (in USD).

Business Purpose: Direct measurement of Average Revenue Per User (ARPU) and customer financial account valuation.

Can it predict churn? Yes (Strong Positive Correlation). High monthly charges (\$70–\$110+) strongly correlate with higher churn rates, especially for Fiber Optic users lacking long-term contract discounts.

20. TotalCharges

NUMERIC FLOAT (CURRENCY)

HIGH IMPORTANCE

PREDICTS: YES (STRONG)

Business Meaning: Cumulative revenue collected from the customer over their entire relationship lifecycle.

Business Purpose: Measures historical Customer Lifetime Value (CLV) and total revenue contribution.

Can it predict churn? Yes (Strong). Total charges combine information from tenure and monthly charges. Low total charges identify either high churn risk (new customers) or low ARPU users. Note: Requires handling missing values caused by spaces in zero-tenure accounts.

21. Churn

CATEGORICAL (BINARY)

TARGET VARIABLE

TARGET OUTPUT

Business Meaning: Target binary label indicating whether the customer churned during the observation window ("Yes", "No").

Business Purpose: Primary KPI for customer retention analysis, predictive modeling targets, and proactive intervention triggers.

Can it predict churn? Target Label. This is the dependent response variable (\$Y\$) used for supervised machine learning training, cross-validation scoring, and ROC-AUC evaluation.

Machine Learning & Feature Engineering Recommendations

- **Data Preprocessing:** Drop `customerID` before training. Convert `TotalCharges` from string to float (handle empty string space values by replacing with median or 0).
- **High-Value Feature Transformations:** Create interaction term `MonthlyCharges / (tenure + 1)` (price sensitivity index) and `Total\_Services\_Subscribed` (count of active security, tech support, backup, and streaming add-ons).
- **Top 5 Drivers of Churn:** Contract Type (Month-to-month), Tenure, Payment Method (Electronic Check), Internet Service (Fiber Optic), and Tech Support / Online Security adoption.
- **Proactive Retention Strategy:** Target single month-to-month fiber optic users during months 1–6 with auto-pay incentives and bundled Tech Support / Online Security discounts to convert them into long-term subscribers.